

WEN-HORNG SHEU

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EDUCATION

PhD in Computer Science at the University of California, Davis; GPA: 4.0/4.0 *2023-Present*

· Research area: Distributed Algorithms, Streaming Algorithms

Master of Computer Science at National Tsing Hua University; GPA: 3.9/4.0 *2019-2021*

· Specialized courses: Approximation Algorithms, Computational Geometry

Bachelor of Computer Science at National Tsing Hua University; GPA: 3.85/4.0 *2015-2019*

· Specialized courses: Advanced Data Structure, Randomized Algorithms, Parallel Algorithm Design

PUBLICATIONS

Following the convention in theoretical computer science, author names are ordered alphabetically unless stated otherwise.

1. **A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic**
with Slobodan Mitrović
SPAA 2025 (ACM Symposium on Parallelism in Algorithms and Architectures)
2. **Faster MPC Algorithms for Approximate Allocation and Matching in Uniformly Sparse Graphs**
with Jakub Łącki, Slobodan Mitrović, and Srikanth Ramachandran
SPAA 2025 (ACM Symposium on Parallelism in Algorithms and Architectures)
3. **Faster Semi-streaming Matchings via Alternating Trees**
with Slobodan Mitrović, Anish Mukherjee, Piotr Sankowski
ICALP 2025 (EATCS International Colloquium on Automata, Languages, and Programming)
4. **Kernelization and Approximation Algorithms for Finding a Perfect Phylogeny from Mixed Tumor Samples**
Wen-Horng Sheu and Bing-Feng Wang (contribution order)
TCBB 2025 (IEEE Transactions on Computational Biology and Bioinformatics)
5. **New Algorithms for Constructing Frequency Difference Consensus Trees**
Bing-Feng Wang, Chih-Yu Li, and Wen-Horng Sheu (contribution order)
TCBB 2025 (IEEE Transactions on Computational Biology and Bioinformatics)
6. **Parameterized Complexity for Finding a Perfect Phylogeny from Mixed Tumor Samples**
Wen-Horng Sheu and Bing-Feng Wang (contribution order)
SIDMA 2023 (SIAM Journal on Discrete Mathematics)

TALKS

- **A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic**
INSAIT, May 2025
- **Toward Optimal Semi-streaming Algorithm for $(1 + \epsilon)$ -approximate Maximum Matching**
Simons Institute for the Theory of Computing, UC Berkeley, Aug 2024
in Workshop on Local Algorithms 2024

RESEARCH EXPERIENCE

Graduate Research Assistant

University of California, Davis

2023 - Present

Davis, CA

- Research area: distributed algorithms, streaming algorithms.
- Studied the maximum matching problem in distributed and streaming settings.

Research Assistant

National Tsing Hua University

2021 - 2023

Hsinchu, Taiwan

- Research area: parameterized algorithms, computational biology.
- Proposed new algorithms for problems that have applications in cancer genomics and phylogenetic analysis.
- Created problems for the International Collegiate Programming Contest (ICPC).

TEACHING EXPERIENCE

Teaching Assistant at University of California, Davis

- Algorithm Design and Analysis (Winter 2025 and Summer Session I 2024)
- Special Topics in Theoretical Computer Science (Winter 2024)

Teaching Assistant at National Tsing Hua University

- Computational Geometry (Spring 2022 and Spring 2020)
- Parallel Algorithm Design (Spring 2022 and Fall 2019)
- Design and Analysis of Algorithms (Fall 2021, Fall 2020, and Fall 2019)

HONORS AND AWARDS

- **Gold Award** in *the 2019 ICPC Asia Pacific Taipei-Hsinchu Regional Contest*
 - Attended several programming contests, including ICPC, with other students in undergraduate years.
 - Built strong abilities in teamwork and problem-solving.
- **Second Place Award** in *the ACM TAU 2018 Contest on Path Reporting*
 - Undergraduate research project: created a tool to perform timing analysis on input circuits.
- **Grandmaster on Codeforces**
 - Ranked as a grandmaster, within top 1% globally on the online competitive programming platform Codeforces (as of Feb 2021).
 - Placed top 1% (out of 10,000+ contestants globally) in four online programming contests on Codeforces.
- **Google Code Jam 2021 Round 3 Qualifier**
 - Competed in the programming contest Google Code Jam 2021.
 - Placed 255-th in Round 3, within top 1% of all 37,000+ participants of the qualification rounds.

ACADEMIC SERVICES

Served as an **external reviewer** for the following conferences and journals

- Conferences: SOSA 2025, SODA 2025, ICALP 2025.
- Journal: Distributed Computing (2025), TALG (2025).

SKILLS

Coding Languages

C, C++, Python

Tools

Git, L^AT_EX, Microsoft Office

Languages

English (fluent), Chinese (native)